



The Arrow Outlasts the Order

A Survival Analysis of Temporary Floor-Marking Wayfinding on a Suburban Shopping Strip

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Abstract. Temporary floor arrows are a familiar fixture of footpath resurfacing works: tape or paint laid down to route pedestrians around a closed section of path while a new surface cures. Municipal guidance treats their removal as incidental to project completion; to our knowledge no study has instrumented that assumption directly. We identify 41 taped arrow-sets laid down for footpath resurfacing jobs across nine shopping strips over an 18-month window, and track each set's continued presence against the jobs' recorded completion dates through monthly site audits. A Kaplan-Meier estimate puts 54% of sets still present 26 weeks after completion and 17% still present at the survey's close, 78 weeks out. Whether a set's originating works order specified a removal deadline does not predict how long it survives (median 214 days with a stated deadline versus 227 days without; Mann-Whitney U, $p=0.71$, n.s.). Semi-structured interviews with 22 shopkeepers adjacent to the surveyed sites find that most cannot identify who would be responsible for removing a set, and several describe treating a still-taped arrow as a legitimate, if unofficial, wayfinding fixture. We situate the finding within a broader pattern in which a measure adopted under a stated, bounded justification persists once that justification lapses and no default removal authority has been assigned.

Introduction

A resurfacing crew closes half a footpath, tapes a run of directional arrows across the remaining width, and the detour holds until the surface cures and the crew returns to lift the tape. The arrangement is temporary by construction: the works order that authorises it specifies a job, not a standing wayfinding scheme, and the tape's only sanctioned purpose is to survive the job. What happens once the job itself is finished, and who is responsible for the tape once the reason for it is gone, has not, to our knowledge, been asked directly of any such arrangement.

The question sits inside a familiar institutional pattern. A measure introduced to meet a bounded, stated condition regularly outlasts that condition once no party is assigned the job of noticing it has lapsed [1], [2]. Behavioural work on defaults and the status quo shows that an arrangement, once in place, acquires an inertial

pull independent of anyone's judgement that it should continue [3], [4], [5]. Tactical-urbanism research documents the same pattern moving in the opposite normative direction: a deliberately temporary intervention becoming a permanent one through exactly the inertia these theories predict, when a city decides afterwards that it likes the result [6]. Floor-arrow tape offers a cleaner test case than either literature usually gets, because nobody involved in laying it down intends it to become permanent, wanted or not; whatever happens to it happens by default rather than by decision.

We survey the arrangement directly. Our contributions are threefold. First, we provide, to our knowledge, the first survival analysis of a temporary floor-marking wayfinding measure against the works order that authorised it, in a non-institutional street setting. Second, we show that the works order's own removal terms

do not predict how long a marking survives, suggesting the document that licenses the tape has little practical bearing on what un-licenses it. Third, drawing on interviews with adjacent shopkeepers, we describe how a still-present arrow-set comes to function as an unofficial wayfinding fixture in its own right, independent of anyone's decision that it should.

Related work

Path dependence supplies the broadest frame: early, often contingent, choices constrain what follows because reversing them costs more than the choice itself ever indicated it might [1]. The concept has been applied to bureaucratic drift in federal programmes [2], to institutional integration in health and social care [7], and, closer to our setting, to adaptation measures whose own persistence can outlast the risk they were built to address [8]. A parallel literature on organisational routines asks why a routine performed flexibly nonetheless persists once its original occasion for existing recedes [9], tracing the mechanism to a routine's dual nature as both an idea and a set of repeated performances that reproduce it without requiring anyone to re-authorise it each time [10], [11]. Status quo bias offers an individual-level mechanism beneath the institutional pattern: once an arrangement is in place, deviating from it is judged and felt differently from arriving at the same arrangement from a blank slate [3], and a default option retains its pull even when the cost of switching away from it is trivial [4], [5]. Tactical-urbanism scholarship documents a mirror case in the same physical medium as ours — a deliberately temporary street marking outliving its trial period, though there by design rather than oversight, once a city opts to keep what the trial produced [6]. Wayfinding-signage research has assessed pedestrians' visual preferences for markings [12], [13], but not, to our knowledge, whether a marking's own presence tracks the paperwork that licensed it. Prior work at this institution has found a campus bike-stall tag outlives the claim that produced it [14], and that a formally "resolved" flag on an exam-hall integrity record persists past the exam period it was raised in far more often than it is closed out [15]; both point to the same absence of an assigned removal authority we test here in a street setting.

Method

Site identification. We obtained the public works register for nine shopping strips across the metropolitan area, each maintained by its local council, and extracted every footpath resurfacing job recorded as complete within an 18-month window. For each job we identified whether the contractor's detour, visible in the job's own site photographs or noted in the works order, included a taped or painted arrow-set directing pedestrians around the closed section; jobs without a documented arrow-set were excluded. This yielded 41 arrow-sets across the nine strips.

Survival measure. Beginning from each job's recorded completion date, we conducted a monthly site visit and photographed the relevant stretch of footpath, coding the arrow-set as present or removed. A set was scored removed at the first visit on which no trace of the original tape or paint remained (a faded but still-continuous line counted as present; an isolated fragment carrying no directional information did not). Sets still present at the survey's close, 78 weeks after the latest job's completion, were right-censored. We estimated the survivor function

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where d_i is the number of sets removed at visit i and n_i is the number still at risk immediately before it [16].

Works-order term. We coded each job's works order for whether it specified a removal deadline for temporary line-marking (a standard clause in some but not all of the nine councils' contract templates) and compared removal-time distributions between the specified- and unspecified-deadline groups with a Mann-Whitney U test.

Removal trigger. For the 34 sets removed within the survey window, we classified the apparent trigger from the site-visit record and, where available, the council's own maintenance log: a routine resurfacing or line-marking audit, unrelated follow-up roadworks disturbing or removing the tape, a shopkeeper request logged with the council, or no reason recorded.

Interviews. We conducted semi-structured interviews with 22 shopkeepers whose frontage

adjoined a surveyed arrow-set (present or removed), asking who they believed was responsible for the marking's upkeep and removal, and how, if at all, they used it. Transcripts were coded using thematic analysis [17].

Results

Across the nine strips, 34 of the 41 surveyed arrow-sets were removed within the 78-week survey window; the remaining seven were still present at its close. The Kaplan-Meier estimate (Figure 1) puts 91% of sets still present eight weeks after their originating job's completion, 54% still present at 26 weeks, 31% at 52 weeks, and 17% — the seven right-censored sets — at 78 weeks. Estimated median survival across the pooled sample is 32 weeks (224 days).

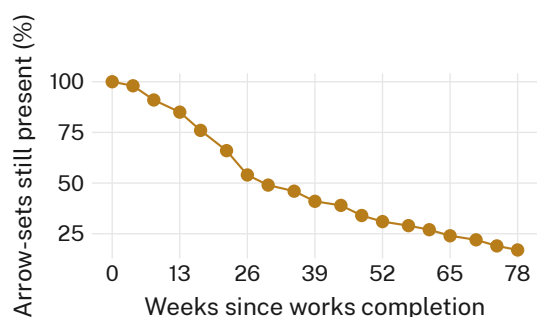


Figure 1: By 26 weeks after their originating job's completion, 54% of taped arrow-sets remained in place; by 78 weeks, 17% (7 of 41) had still not been removed, with no scheduled removal recorded for any of them.

Whether the originating works order specified a removal deadline for temporary line-marking does not predict how quickly a set came down (Figure 2): sets with a stated deadline ($n=19$) show a median removal time of 214 days, against 227 days for sets with no stated deadline ($n=22$; Mann-Whitney $U=193$, $p\text{-value}=0.71$, not significant). The clause exists in a minority of the nine councils' works-order templates and, on this evidence, has no measurable bearing on what happens once the job is done.

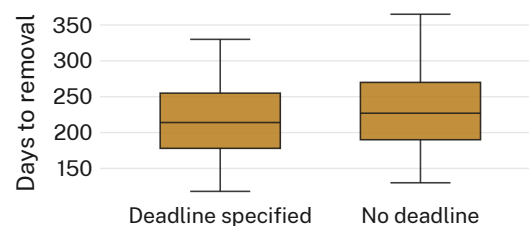


Figure 2: Arrow-sets whose works order specified a removal deadline took a median 214 days to come down ($n=19$); sets with no stated deadline took a median 227 days ($n=22$, not significant).

Among the 34 sets removed within the window, the apparent trigger was a routine resurfacing or line-marking audit in 9 cases, unrelated follow-up roadworks disturbing or removing the tape in 14, a shopkeeper request logged with the council in 4, and no reason recorded in the remaining 7 (Figure 3) — in under a third of removals, in other words, did anyone act because the arrow-set itself was the thing under review.

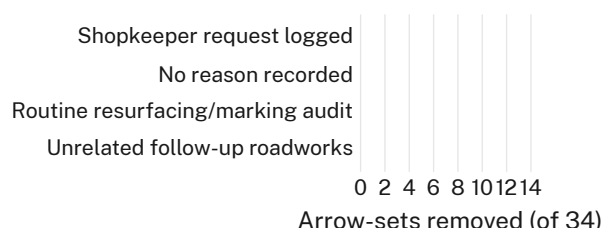


Figure 3: Of the 34 arrow-sets removed within the survey window, an unrelated follow-up roadworks visit disturbed or removed the tape in 14 cases — more than any other recorded trigger, including a scheduled audit of the marking itself (9).

Of the 22 shopkeepers interviewed, 17 (77%) could not name a person, role, or council unit they believed was responsible for removing a marking outside a scheduled audit, several noting that they had never previously considered the question. Six (27%) described treating a still-present arrow-set as a legitimate, if unofficial, wayfinding aid for customers, independent of whether the underlying footpath closure it once guided around still existed.

Discussion

The pattern across all three measures points the same way: an arrow-set's survival tracks something other than the paperwork that authorised it, and in most cases something other than any deliberate decision about the arrow-set at all. Removal happens, on this evidence, most often

as a side effect of unrelated works returning to the same stretch of footpath, and least often because anyone identified the arrow-set as itself due for review. This sits comfortably alongside organisational-routines research showing that a routine's persistence needn't — and often doesn't — depend on anyone re-authorising it each time it recurs [9], [10], and alongside status-quo-bias findings that an arrangement, once installed, need not clear any bar higher than nobody happening to remove it [3]. The shopkeeper interviews suggest a further mechanism specific to this setting: once a marking has been present long enough, a subset of the people who live beside it begin treating it as a fixture rather than a leftover, which raises, however modestly, the local cost of anyone deciding to peel it up. We read the absence of a works-order effect as consistent with a simpler explanation than any of the above: no document in our sample assigns removal to anyone by default, and in that absence, the tape holds until something else disturbs it.

Limitations

Our sample covers nine shopping strips within a single metropolitan area and a single 18-month window; we cannot rule out that a council with an unusually well-resourced maintenance unit would show a different pattern, though the consistency of the null works-order effect across all nine strips' differing contract templates suggests the finding does not depend on any one council's practice. Monthly site visits cannot date a removal more precisely than the visiting interval, which coarsens our survival estimate without, we expect, biasing it in either direction. The shopkeeper sample is drawn from frontages adjacent to a surveyed arrow-set and may under-represent residents and passers-by who notice a marking without working beside it daily.

Conclusion

A floor arrow taped down to route pedestrians around a footpath closure is, on this evidence, no more likely to come down because the works order said it should than because nothing else happened to disturb it first. The document that licenses a temporary measure does not, in this setting, carry any authority over un-licensing it, and in that gap a marking with no one assigned to remove it tends simply to remain. Future work might test whether adding a named re-

moval owner to the works-order template — the same intervention organisational-routines and default-effects research would predict should matter — actually shortens an arrow-set's survival, or whether, like the deadline clause already in some templates, it turns out to make no difference at all.

References

- [1] J. Beyer, "On a Branching Route: The Spectrum of Path Dependence in Policy Research," *Review of Policy Research*, 2026, doi: [10.1111/ropr.70007](https://doi.org/10.1111/ropr.70007).
- [2] R. Mukherji and S. H. Zarhani, "Policy Paradigms and Path Dependence: The Endogenous Roots of Institutional Displacement and Drift in India," *Global Public Policy and Governance*, 2021, doi: [10.1007/s43508-021-00007-w](https://doi.org/10.1007/s43508-021-00007-w).
- [3] W. Samuelson and R. Zeckhauser, "Status Quo Bias in Decision Making," *Journal of Risk and Uncertainty*, 1988, doi: [10.1007/BF00055564](https://doi.org/10.1007/BF00055564).
- [4] R. H. Thaler and C. R. Sunstein, "Libertarian Paternalism," *American Economic Review*, 2003, doi: [10.1257/000282803321947001](https://doi.org/10.1257/000282803321947001).
- [5] E. J. Johnson and D. Goldstein, "Do Defaults Save Lives?," *Science*, 2003, doi: [10.1126/science.1091721](https://doi.org/10.1126/science.1091721).
- [6] N. Ortar and P. Rérat, *Cycling Through the Pandemic: Tactical Urbanism and the Implementation of Pop-Up Bike Lanes in the Time of COVID-19*. Springer, 2024. doi: [10.1007/978-3-031-45308-3](https://doi.org/10.1007/978-3-031-45308-3).
- [7] H. Trouvé, Y. Couturier, F. Etheridge, O. Saint-Jean, and D. Somme, "The Path Dependency Theory: Analytical Framework to Study Institutional Integration. The Case of France," *International Journal of Integrated Care*, 2010, doi: [10.5334/ijic.544](https://doi.org/10.5334/ijic.544).
- [8] S. Hanger-Kopp, T. Thaler, S. Seebauer, T. Schinko, and C. Clar, "Defining and Operationalizing Path Dependency for the Development and Monitoring of Adaptation Pathways," *Global Environmental Change*, 2022, doi: [10.1016/j.gloenvcha.2021.102425](https://doi.org/10.1016/j.gloenvcha.2021.102425).
- [9] J. A. Howard-Grenville, "The Persistence of Flexible Organizational Routines: The Role of Agency and Organizational Context,"

- Organization Science*, 2005, doi: [10.1287/orsc.1050.0150](https://doi.org/10.1287/orsc.1050.0150).
- [10] M. S. Feldman, “Organizational Routines as a Source of Continuous Change,” *Organization Science*, 2000, doi: [10.1287/orsc.11.6.611.12529](https://doi.org/10.1287/orsc.11.6.611.12529).
 - [11] M. D. Cohen and P. Bacdayan, “Organizational Routines Are Stored as Procedural Memory: Evidence from a Laboratory Study,” *Organization Science*, 1994, doi: [10.1287/orsc.5.4.554](https://doi.org/10.1287/orsc.5.4.554).
 - [12] J. Zhou and N. Ujang, “An Analysis of Pedestrian Preferences for Wayfinding Signage in Urban Settings: Evidence from Nanning, China,” *Buildings*, 2024, doi: [10.3390/buildings14092986](https://doi.org/10.3390/buildings14092986).
 - [13] J. Zhou, N. Ujang, M. S. Abd Manan, and F. Abdul Aziz, “Bridging Perceptual Gaps: Designers vs. Non-Designers in Urban Wayfinding Signage Preferences,” *Sustainability*, 2024, doi: [10.3390/su16229653](https://doi.org/10.3390/su16229653).
 - [14] R. Sabel, M. Hanke, and B. Ntuli, “The Rack Forgets Slower Than the Rider: Cataloguing Why a Campus Bike-Stall Tag Outlives Its Claim,” *Slop University technical report*, 2026, doi: [10.5555/slop.etwfmk](https://doi.org/10.5555/slop.etwfmk).
 - [15] M. Solheim and R. Sabel, “Closed, Reclassified, or Left Open: A Hand-Coded Taxonomy of Bathroom-Break Integrity-Attentiveness Flags Across Eight Terms at the Exit Sign-Out Table,” *Slop University technical report*, 2026, doi: [10.5555/slop.jw4nw9](https://doi.org/10.5555/slop.jw4nw9).
 - [16] E. L. Kaplan and P. Meier, “Non-parametric Estimation from Incomplete Observations,” *Journal of the American Statistical Association*, 1958, doi: [10.1080/01621459.1958.10501452](https://doi.org/10.1080/01621459.1958.10501452).
 - [17] V. Braun and V. Clarke, “Using Thematic Analysis in Psychology,” *Qualitative Research in Psychology*, 2006, doi: [10.1191/1478088706qp063oa](https://doi.org/10.1191/1478088706qp063oa).